

G8-02-CX-pressure-strain-nonlocal-kill-test-v1: $\nabla^2 p$ gain over W_+

ZFL-CX

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1 Frozen input

$$\boxed{G7 = \text{FROZEN}, G8\text{-DESIGN-AUDIT} = \text{DONE}, G8\text{-01} = \text{KILLED}} \quad \boxed{\mathcal{C}_{\text{PROVEN}} = \{C_E^{\text{weak}}, C_{L^3}, C_F^{\text{conc}}, C_I^{\text{freq}}, C_J^{\text{m}}\}}$$

$$\boxed{\mathcal{C}_{\text{PROVEN}} \Rightarrow \perp \text{ OPEN}} \quad \boxed{W_+(t) = \left(\int \omega^T S \omega \right)_+, \int_0^{T^*} W_+ = +\infty \text{ if } T^* < \infty}$$

2 Mission G8-02

Mechanism: $D_t S = -S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S$. Audit only $\boxed{\nabla^2 p}$ effect on W . Target A: $W_+ \leq F$, $\int_0^{T^*} F < \infty$ Target B: $\mathcal{C}_{\text{PROVEN}} \wedge C_{G8-02} \Rightarrow \perp$ Otherwise KILLED.

3 Structure of pressure

$-\Delta p = \partial_i \partial_j (u_i u_j) = \text{tr } S^2 - \frac{1}{2} |\omega|^2 = \text{tr}(S^2 - \Omega^2)$. Hence $p = \Delta^{-1} \text{tr}(S^2 - \Omega^2)$, $\nabla^2 p = \mathcal{R}[S^2 - \Omega^2]$ with $\mathcal{R}$ matrix Riesz transform, bounded on L^q , $1 < q < \infty$. So $\boxed{\|\nabla^2 p\|_{L^q} \lesssim \|S^2 - \Omega^2\|_{L^q}}$.

4 Barriers

1. If only $\|\nabla^2 p\|_q \lesssim \|S^2 - \Omega^2\|_q \Rightarrow$ KILLED. 2. If Calderon-Zygmund returns same critical scale $\Rightarrow$ KILLED. 3. If assumes regularity to prove regularity $\Rightarrow$ KILLED. 4. If nonlocality gives no cancellation/coercivity on $W_+ \Rightarrow$ KILLED. 5. No $W_+ \lesssim \|u\|_3 \|\nabla \omega\|_2^2$ as bridge (G8-01).

5 Test: does $\nabla^2 p$ give integrable gain on W_+ ?

We look for estimate $W_+ \leq F(\|\nabla^2 p\|)$. $W = \int \omega^T S \omega$. Evolution of W via $D_t S$: $\frac{d}{dt} W$ contains $-\int \omega^T \nabla^2 p \omega$. But $\int \omega^T \nabla^2 p \omega = \int p_{ij} \omega_i \omega_j$.

Write $\int \omega_i \omega_j \nabla_i \nabla_j p = -\int \partial_j (\omega_i \omega_j) \partial_i p$ up to boundary. No sign: p not signed, $\nabla^2 p$ not coercive.

Nonlocal representation: $\nabla^2 p(x) = PV \int K(x-y)(S^2 - \Omega^2)(y) dy$. K is L^1 -singular, zero mean, no L^∞ gain. Under $C_F^{\text{conc}} + C_I^{\text{freq}}$, $S^2 - \Omega^2$ concentrates at $r_n \rightarrow 0$ with $L^{3/2}$ -scale critical. Riesz transform preserves critical scale: $\|\nabla^2 p\|_{L^{3/2}} \sim \|S\|_{L^3}^2$.

Thus any bound $W_+ \lesssim \|\nabla^2 p\|_{L^{3/2}} \|\omega\|_{L^3}^2$ or $W_+ \lesssim \|\nabla^2 p\|_{L^q} \|\omega\|_{L^{2q}}^2$ reduces again to $\|u\|_{L^3} \|\nabla \omega\|_2^2$ scaling after Sobolev. Formally $F(t) \sim \|S(t)\|_{L^3}^3$ or $\|u\|_{L^3}^a \|\nabla \omega\|_2^b$, with $a + b = 3$, exactly the dimensional class forbidden as bridge. Since C_{L^3} allows $\|u\|_3 \rightarrow \infty$, $\int_0^{T^*} F = +\infty$ compatible with $\mathcal{C}_{\text{PROVEN}}$.

No cancellation: trace-free gives $\int \text{tr} \nabla^2 p = \int \Delta p$, but W uses anisotropic $\xi^T \nabla^2 p \xi$, $\xi = \omega/|\omega|$. C_J^{neg} already encodes that alignment is not controlled. Nonlocality does not improve alignment; kernel K is even, does not create negativity for $\xi^T \nabla^2 p \xi$.

Pressure as Lagrange multiplier enforces incompressibility, not damping of W_+ : $D_t S$ term $-\nabla^2 p$ is needed to keep $\text{tr} S = 0$, not to produce $W_+ \leq 0$.

Therefore no quantitative gain $\boxed{W_+ \leq F, \int F < \infty}$ emerges from pure $\nabla^2 p$ structure.

6 Outcome

KILLED

- Only $\|\nabla^2 p\|_q \lesssim \|S^2 - \Omega^2\|_q$ obtained, same critical scale.
- Calderon-Zygmund preserves concentration scale of C_F^{conc}, C_I^{freq} .
- No coercivity/cancellation on W_+ from nonlocality.
- Any F constructed is in forbidden dimensional class, not integrable under $\mathcal{C}_{\text{PROVEN}}$.

$\mathcal{C}_{\text{PROVEN}} \not\perp$ remains OPEN. G8-01 and G8-02 both exhaust stretching algebra and pressure-strain without integrable bridge.